

LECTURE 9: SAMPLE SURVEYS

Key concepts:

- Complete data model
- Inclusion model
- Complete and observed data likelihood
- Ignorability
- Missing at random
- Distinct parameters
- Finite and super-population inference

1. INTRODUCTION

2. MODELS FOR DATA COLLECTION

The following notation and terminology is from Rubin, 1976:

Consider a finite population $U = \{1, \dots, N\}$ of size N . The population data is $(y_1, \dots, y_N)$ (where each y_i may be a vector $(y_{i1}, \dots, y_{iJ})$ of dimension J , so the population data can be viewed as a matrix y of dimension $N \times J$ with (i, j) th element y_{ij}). Let I be a matrix also of dimension $N \times J$ where $I_{ij} = 1$ if y_{ij} is observed and $I_{ij} = 0$ if y_{ij} is not observed (“missing”). This accommodates both unit non-response ($I_{ij} = 0$ for $j = 1, \dots, J$) and item non-response ($I_{ij} = 0$ only for some j). We assume that I_{ij} is observed for all i and j . Let ‘obs’ = $\{(i, j) : I_{ij} = 1\}$ and ‘mis’ = $\{(i, j) : I_{ij} = 0\}$ be the sets of indices that are observed and missing, respectively. We will use y_{obs} and y_{mis} denote the observed and missing components of y , respectively.

The above assumes (implicitly) *stability* (or, sometimes, *SUTVA*, the *stable unit treatment value assumption*), which is that the unobserved values of y are unaffected by which units are observed or not. This is generally not considered a problem in surveys, where interference among units is unlikely to occur.

In surveys, we often have *auxiliary variables* or *covariates* x , which are available for the entire population. (Often it suffices to have the distribution of the covariates available, so we don’t need the micro data.)

The *complete data model* is

$$p(y, I | \theta, \phi) = p(y | \theta) p(I | y, \phi)$$

where $p(y | \theta)$ is the *complete data likelihood* and $p(I | y, \phi)$ is the *inclusion model*. Here, θ are the parameters of the complete data model and usually of primary interest, while ϕ are the parameters of the inclusion model. At this point, they may overlap, so we haven’t made any substantive assumptions in this factorization (but the assumptions will come shortly).

If there are covariates x available, we can condition upon them to obtain

$$p(y, I|\theta, \phi) = p(y|x, \theta) p(I|x, y, \phi).$$

This will be the form of the likelihood that we will work with. All inferences will be made conditional upon the covariates.

The *observed data likelihood* (conditional upon x) is

$$p(y_{\text{obs}}, I|x, \theta, \phi) = \int p(y, I|x, \theta, \phi) dy_{\text{mis}}.$$

The joint posterior of (θ, ϕ) given the observed data (y_{obs}, I) (and implicitly x) is

$$\begin{aligned} p(\theta, \phi|x, y_{\text{obs}}, I) &\propto p(\theta, \phi|x) p(y_{\text{obs}}, I|x, \theta, \phi) \\ &= p(\theta, \phi|x) \int p(y, I|x, \theta, \phi) dy_{\text{mis}} \\ &= p(\theta, \phi|x) \int p(y|x, \theta) p(I|x, y, \phi) dy_{\text{mis}}. \end{aligned}$$

The marginal posterior distribution of θ is

$$\begin{aligned} p(\theta|x, y_{\text{obs}}, I) &= \int p(\theta, \phi|x, y_{\text{obs}}, I) d\phi \\ &\propto \int p(\theta, \phi|x) \int p(y|x, \theta) p(I|x, y, \phi) dy_{\text{mis}} d\phi \\ &= p(\theta|x) \int p(\phi|x, \theta) \int p(y|x, \theta) p(I|x, y, \phi) dy_{\text{mis}} d\phi. \end{aligned}$$

Alternatively, we could make inferences “ignoring” the data collection process, using

$$\begin{aligned} p(\theta|x, y_{\text{mis}}) &\propto p(\theta|x) p(y_{\text{obs}}|x, \theta) \\ &= p(\theta|x) \int p(y|x, \theta) dy_{\text{mis}}. \end{aligned}$$

We will obtain the same inferences if

$$p(\theta|x, y_{\text{mis}}) \propto p(\theta|x, y_{\text{obs}}, I).$$

We say that the missing data mechanism is *ignorable* if this condition holds. This terminology is confusing, since the posterior distribution in either case involves the inclusion model.

Next, we investigate conditions under which a missing data mechanism is ignorable. Data are said to be *missing completely at random* (MCAR) if

$$p(I|x, y, \phi) = p(I|\phi).$$

This condition is satisfied by simple random sampling without replacement (SRSWOR) where

$$p(I|\phi) = \binom{N}{n}^{-1} \quad \text{if } \sum_{i=1}^N I_i = n,$$

which does not depend on x , y , or even ϕ . A weaker assumption is *missing at random* (MAR) where

$$p(I|x, y, \phi) = p(I|x, y_{\text{obs}}, \phi).$$

This is satisfied by stratified random sampling if x includes the stratum indicators. See below for more discussion.

One other condition that we will use is *distinct parameters*, where θ and ϕ are independent in the prior distribution:

$$p(\theta, \phi|x) = p(\theta|x) p(\phi|x).$$

Proposition 1. *If the data are missing at random and the parameters are distinct, then the design is ignorable.*

Proof. Using distinct parameters and MCAR,

$$\begin{aligned} p(\theta|x, y_{\text{obs}}, I) &\propto p(\theta|x) \int \int p(\phi|x) p(y|x, \theta) p(I|x, y_{\text{obs}}, \phi) dy_{\text{mis}} d\phi \\ &= p(\theta|x) \int p(\phi|x) p(I|x, y_{\text{obs}}, \phi) \int p(y|x, \theta) dy_{\text{mis}} d\phi \\ &= p(\theta|x) p(y_{\text{obs}}|x, \theta) \int p(\phi|x) p(I|x, y_{\text{obs}}, \phi) d\phi \\ &\propto p(\theta|x) p(y_{\text{obs}}|x, \theta) \propto p(\theta|x, y_{\text{obs}}). \end{aligned}$$

□

3. FINITE POPULATION AND SUPERPOPULATION INFERENCE

Bayesian and model-based inferences in finite population sampling are based on a *superpopulation model*. In model-based inference, this is a data generating process for the population data. In Bayesian inference, it is a prior distribution on the finite population parameters.

The simplest superpopulation model is

$$y_1, \dots, y_N \stackrel{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$$

so $\theta = (\mu, \sigma^2)$. We assume SRSWOR so

$$p(I) = \begin{cases} \binom{N}{n} & \text{if } \sum_{i=1}^N I_i = n, \\ 0 & \text{otherwise,} \end{cases}$$

so the missing data models contains no unknown parameters or covariates. The observed data $y_{\text{mis}} = \{y_i : I_i = 1\}$ consist of the n observed values, which may be taken to be $(y_1, \dots, y_n)$ without loss of generality.

First, suppose σ^2 is known, since the ideas emerge more clearly in this case. Suppose $p(\mu) \propto 1$, so the posterior is

$$\mu|y_{\text{obs}} \sim \text{Normal}(\bar{y}_{\text{obs}}, \sigma^2/n),$$

where $\bar{y}_{\text{obs}} = \sum_{i=1}^n y_i/n$. The predictive distribution of $\bar{y}_{\text{mis}} = \sum_{i=n+1}^N y_i/(N-n)$ conditional upon $\bar{y}_{\text{obs}}$ is

$$\bar{y}_{\text{mis}}|\bar{y}_{\text{obs}} \stackrel{\text{iid}}{\sim} \text{Normal}(\bar{y}_{\text{obs}}, N\sigma^2/(N-n)).$$

using

$$\begin{aligned}\mathbb{V}(\bar{y}_{\text{mis}}|y_{\text{obs}}) &= \mathbb{E}[\mathbb{V}(\bar{y}_{\text{mis}}|\mu)|y_{\text{obs}}] + \mathbb{V}[\mathbb{E}(\bar{y}_{\text{mis}}|\mu)|y_{\text{obs}}] \\ &= \mathbb{E}\left[\frac{\sigma^2}{N-n}\middle|y_{\text{obs}}\right] + \mathbb{V}[\mu|y_{\text{obs}}] \\ &= \frac{\sigma^2}{N-n} + \frac{\sigma^2}{n} = \frac{N}{N-n}\sigma^2.\end{aligned}$$

Finally, since the population mean is

$$\bar{y}_U = \frac{n\bar{y}_{\text{obs}} + (N-n)\bar{y}_{\text{mis}}}{N},$$

the predictive distribution of $\bar{y}_U$ given y_{obs} is

$$\bar{y}_U|y_{\text{obs}} \sim \text{Normal}\left(\bar{y}_{\text{obs}}, \left(\frac{N-n}{N}\right)\frac{\sigma^2}{n}\right),$$

similar to the conventional frequentist sampling distribution, except with the roles of $\bar{y}_U$ and $\bar{y}_{\text{obs}}$ reversed. The normality of the sampling distribution has only asymptotic justification.

A closely related result holds if σ^2 is unknown, but we have the improper joint prior

$$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2}.$$

The posterior distribution is

$$\mu, \sigma^2|y_{\text{obs}} \sim \text{Normal-Inverse } \chi^2(\bar{y}_{\text{obs}}, s^2, n-1).$$

The predictive distribution of $\bar{y}_U$ conditional upon y_{obs} is

$$\bar{y}_U|y_{\text{obs}} \sim t\left(\bar{y}_{\text{obs}}, \left(\frac{N-n}{N}\right)\frac{s^2}{n}, n-1\right).$$

In design-based inference, it is shown that these are unbiased estimates for the finite population mean and the variance of the sample mean as an estimator of the population mean. It is, however, impossible to obtain the t distribution from design-based assumptions, though it is often used anyway with some hand waving.

4. CLUSTER SAMPLING

In most situations, it is difficult to draw a SRSWOR, so we resort to multistage sampling by first selecting clusters, enumerating the selected clusters, and then subsampling these using SR-SWOR (or further stages of subsampling). In *BDA3*, pp. 211–212, a cluster sample of Australian schoolchildren is given. In the first stage, 72 schools were selected. From each selected school, two classes were selected and all children in the selected classes were asked whether they walked to school. This is actually three stages of sampling, though at the final stage all units were selected.

Following the notation in *BDA3*, index children by i , classes by j , and schools by k , where $i = 1, \dots, N_{jk}$, $j = 1, \dots, M_k$, and $k = 1, \dots, K$. The total number of children in school k

is $N_k = \sum_{j=1}^{M_k} N_{jk}$ and the number of children in the city (Melbourne) is $N = \sum_{k=1}^K N_k$. The proportion of children who walk to school in class j in school k and in all of school k are

$$\bar{y}_{\cdot jk} = \frac{1}{N_{jk}} \sum_{i=1}^{N_{jk}} y_{ijk} \quad \bar{y}_{\cdot \cdot k} = \frac{1}{N_k} \sum_{j=1}^{M_k} N_{jk} \bar{y}_{\cdot jk}$$

The estimand of interest is the proportion of children in all of Melbourne who walk to school,

$$\bar{y}_U = \frac{1}{N} \sum_{k=1}^K N_k \bar{y}_{\cdot \cdot k}$$

Gelman *et al.* use the model

$$\bar{y}_{\cdot jk} \sim \text{Normal}(\theta_{jk}, \sigma_{jk}^2)$$

where

$$\sigma_{jk}^2 = \frac{\bar{y}_{\cdot jk}(1 - \bar{y}_{\cdot jk})}{N_{jk}}$$

(The variance assumption is weird. It seems like $\sigma_{jk}^2 = \theta_{jk}(1 - \theta_{jk})/N_{jk}$ would make more sense.) They then adopt the following hierarchical model:

$$\begin{aligned} \theta_{jk} &\sim \text{Normal}(\theta_k, \tau_{\text{class}}^2) \\ \theta_k &\sim \text{Normal}(\alpha + \beta M_k, \tau_{\text{school}}^2) \end{aligned}$$

so a correlation between school size and walking to school is permitted. (For example, smaller schools could serve smaller neighborhoods and, hence, be easier to walk to. Or smaller schools could be located in less dense areas which are harder to walk to. We don't need to choose one over the other, but allow the model to flexibly incorporate whatever pattern exists in the data.)

We find the posterior distribution of the population mean by drawing $\tilde{y}_{ijk} \sim \text{Normal}(\theta_{jk}, \tau_{\text{class}}^2)$ for the unselected classes in selected schools, using values of the parameters from the posterior simulations. Similarly, for unselected schools, we simulate $\tilde{y}_{ijk}$ by first drawing $\tilde{\theta}_k$ from $\text{Normal}(\alpha + \beta M_k, \tau_{\text{school}}^2)$ (using posterior simulations of α , β , and τ_{school}^2), then draw $\tilde{\theta}_{jk}$ from $\text{Normal}(\tilde{\theta}_k, \tau_{\text{class}}^2)$ and $\tilde{y}_{ijk}$ from $\text{Normal}(\tilde{\theta}_{jk}, \sigma_{jk}^2)$ (again using posterior simulations for the parameters). Having simulated the entire population, we then compute the population mean for each simulation. Usually, a relatively small number of simulations is adequate.